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Subject: Learning and Ethics of Research

Assignment: Week 9

A. What Is (Probably) Wrong with Nishiura's Team's Paper

1. The analysis did not include severe cases and deaths due to data limitations.
2. The study focused on Tokyo and did not assess geographic heterogeneity in the impact of vaccination.
3. The emergence of new sub lineages and age-dependent differences in the immune response may complicate future assessments.
4. "The population-level impact by vaccine dose is illustrated in Fig. 2. The total impact was estimated at approximately 8.5–9.0 million infections averted by the end of May 2022. The direct effects differed between the programs; 2.6 million infections were prevented by the primary series plus booster program, and 0.6 million infections were prevented by the booster program alone. The indirect impact was obtained by subtracting the direct effect from the total effect, and the proportion of infections indirectly prevented was estimated to be a 70% and 93% total risk reduction owing to the primary series plus booster program and the booster program alone, respectively.."… This is a very large percentage of reduction, however the numbers might be suspicious (too large) because had there been a research where the reduction was actually 93%, the research would have been viral, however, it is not.
5. In the section of "Vaccination Coverage":

"Because the discrepancy between the coverage of the first and second doses was small (0.6% as of June 1, 2022) according to the VRS, we assumed that all individuals who received a first dose subsequently received a second dose." : But what if they don't? It's important to consider uncertainties and not simply to assume.

2. One example of data mishandling by a scientist is the case of Bengü Sezen, a graduate student at Columbia University, who was found to have falsified data in her research for over a decade by editing NMR data to fit her desired results. At least six of her research papers have been withdrawn, and Columbia University has moved to revoke her Ph.D. This incident is a clear example of data misuse in the scientific community.